

RESEARCH ARTICLE

Patterns of Mortality and Factors Influencing Survival of a Recently Restored Elk Population in Ontario, Canada

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Abstract

This study examined patterns of mortality and determinants of survival among elk recently restored to four sites in Ontario, Canada (1998–2005). We predicted that: (1) elk located in release sites closer to the core of their historic range would have higher survival; (2) survival would increase as an animal's time and experience on the landscape increased; and (3) survival rates would decline as animals moved farther away from the release site. During the study, 443 elk were radiocollared and released; 218 mortalities were documented. Predation by wolves was the most important proximate cause of mortality, followed by death due to injuries from translocation and/or capture myopathy, accidents, emaciation, poaching, and *Parelaphostrongylus tenuis* infection. Overall, annual survival of elk across Ontario ranged from 0.45 (0.37–0.53) to 0.81 (0.66–0.90),

with rates being lowest in the years immediately following release and highest in the final years of the study; this pattern was due to high initial mortality from translocation injuries and/or capture myopathy and possibly lack of familiarity with novel habitat. Model-averaged hazards further support this finding, as the most important factor influencing elk survival was the length of holding period, with elk released after limited holding being less likely to survive than those held for longer periods. Our results suggest that mortalities caused by capture myopathy and transportation-related injuries are important sources of risk for translocated elk. The method of introduction to the novel landscape and behavior in the first year should be accommodated via soft-release and appropriate release areas.

Key words: *Cervus elaphus*, elk, mortality, Ontario, restoration, survival.

Introduction

Mechanisms affecting the variation in survival of large ungulates are relatively well understood. Many studies have shown that patterns of numerical change are fuelled by a combination of stochastic environmental variation and density-dependent mortality (Sæther 1997, Gaillard et al. 1998, Coulson et al. 2000). In turn, ungulate mortality is often driven by a variety of factors, including food availability and habitat quality (Bender et al. 2007), climate (Garrott et al. 2003), disease and parasites (Murray et al. 2006), predation (Kunkel & Pletscher 1999), population density (Stewart et al. 2005), and harvest (Ballard et al. 2000), and each has been shown, at least to some degree, to account for annual variability in survival rates.

The restoration of wildlife to areas from which they have been extirpated provides a rare opportunity to study the factors that influence survival in colonizing populations, which may differ from those that influence established populations. A common feature of many wildlife restoration attempts is a high loss of animals immediately following release (Griffith et al. 1989). While persistence largely depends on good habitat conditions, the establishment of a new group of individuals depends on the short-term local survival of released individuals (Armstrong & Seddon 2007). Short-term post-release survival is thus a crucial parameter for measuring the success of new and restored populations.

Recently released animals may experience variable survival rates depending on the age of individuals and cohort size, where an overall increase in survival is related to the total number of animals released and, in particular, the number of animals released in cohorts with traditionally higher survival rates (i.e. adult females) (Sarrazin & Legendre 2000). Furthermore, animals released into good quality habitat in the core of their historical distribution have a greater chance of establishing a self-sustaining population than those released on the periphery (Griffith et al. 1989). Social and ecological familiarity with the landscape may also encourage higher survival as animals are able to disperse and exploit good quality habitat (Stamps

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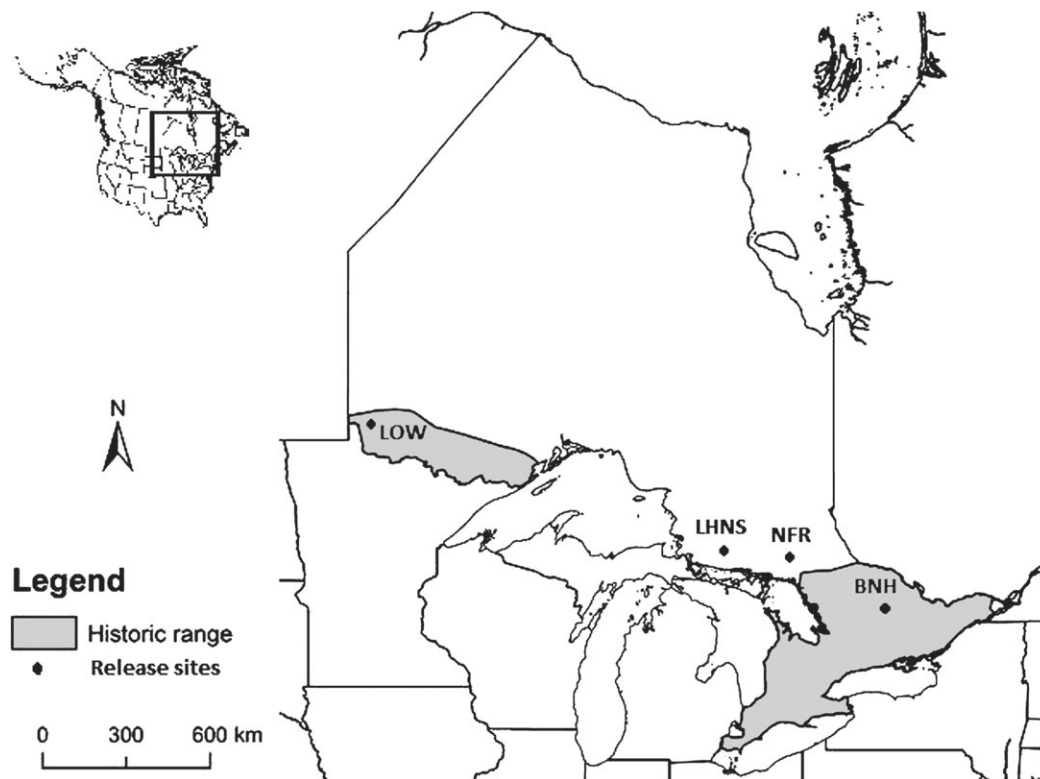


Figure 1. Map depicting the historical distribution of elk in Ontario, Canada (adapted from Peterson 1966) and the four elk release sites in Ontario, Canada 1998–2001. LOW = Lake of the Woods; LHNS = Lake Huron-North Shore; NFR = Nipissing-French River; BNH = Bancroft-North Hastings.

& Swaisgood 2007). Pre- and post-release experiences may be similarly important, as the ability of translocated animals to cope with novel environments has been related to experience with both habitat and mortality risks (Frair et al. 2007). Finally, factors such as release site fidelity, dispersal behavior, and soft versus hard release methodology may be related to social and ecological familiarity with the landscape and can also play a critical role in the variation of survival (Ryckman et al. 2007, Haydon et al. 2008).

Once native to Ontario, elk (*Cervus elaphus*) occupied the deciduous forest westward to 95° longitude, northward to the 47th parallel, and southward to the 34th parallel (O’Gara & Dundas 2002) (Fig. 1). However, demands for meat and agricultural land resulted in the extirpation of these elk during the late 1800s and early 1900s (Witmer 1990). There have been several previous attempts to reintroduce elk to the province, mostly notably in the 1930s when elk were reintroduced to several locations in central Ontario. However, due to concerns regarding the transmission of the giant liver fluke, *Fascioloides magna*, to livestock, most of these animals were destroyed (Addison 1997). Two small remnant populations managed to survive in the Nipissing-French River (NFR) area of Ontario, and in 1996 they were estimated to number ~60 animals (Bellhouse & Broadfoot 1998).

In 1997, the Government of Ontario announced an elk restoration initiative, and from 1998 to 2001 a total of 460 elk were translocated from Elk Island National Park (EINP),

Alberta and released in Ontario. We present results from an analysis of causes of mortality and determinants of survival among elk released into Ontario, Canada, across four distinct release sites; three release areas are on the periphery of the historical range of elk in Ontario. We predicted that: (1) release sites would differ in terms of survival rates experienced by locally released elk, with those animals released farther from the core of their historic range experiencing increased risk due to marginal habitat and higher predator densities; (2) survival rate would be largely influenced by an animal’s experience or familiarity with the Ontario landscape, with survival increasing as animals became more familiar with novel habitat and mortality risks; and (3) variability in survival rates would be influenced by distance moved from the release site, with animals moving farther from the release site having increased risk compared with those remaining close to the release site. We consider that assessment of mortality determinants in restored elk in Ontario will be relevant to understanding the status and prognosis of recovering ungulate populations in the periphery of their native range, as well as inform on the general development of successful methodology for ungulate recovery programs.

Methods

Study Area

The four elk restoration sites extend through Ontario, from Lake of the Woods (LOW; 49°15’N, 93°43’W) in the northwest,

Lake Huron-North Shore (LHNS; 46°26'N, 83°6'W) and Nipissing-French River (NFR; 46°12' N, 80°50'W) in the central part of the province, and Bancroft-North Hastings (BNH; 45°02'N, 83°29'W) in southeastern Ontario (Fig. 1). Three of the release sites, LOW, NFR, and LHNS are located in the northern periphery of the historic range of elk in Ontario, whereas the BNH release is considered to be within the core historic range (Fig. 1). Pair-wise distance between release sites ranges from 250 to 1200 km; therefore, populations at each site are considered distinct.

Climate variability among release areas includes a latitudinal temperature gradient, as well as a precipitation gradient from northwest to southeast (Environment Canada 2003). Mean maximum snow depth in all elk release areas averages between 50 and 65 cm annually (Environment Canada 2003); however, spring green-up generally occurs 1 month earlier (mid to late April) in the more southern elk release site (i.e. BNH) (Hutchinson et al. 1997). Potential competitors to elk in the areas include white-tailed deer (*Odocoileus virginianus*) and moose (*Alces alces*), whereas predators include wolves (*Canis lupus*, *Canis lycaon*), coyotes (*Canis latrans*), and black bears (*Ursus americanus*). Deer density is higher in the south (Hutchinson et al. 1997), while the densities of wolves (~3 per 100 km²) and black bears (40–60 per 100 km²) is higher in the NFR and LOW areas (Rosatte et al. 2007). Finally, human densities vary among the release areas, with relatively low densities in the LOW, NFR, and LHNS release sites (1 person per 1.6 km²), compared with much higher densities (15–19 persons per 1.6 km²) in the BNH region (Hutchinson et al. 1997).

Animal Capture and Monitoring

From 1998 to 2001, 460 elk were transported from Elk Island National Park, Alberta, to four release sites in Ontario (Rosatte et al. 2007). On arrival at Ontario, elk were held in large enclosures to recuperate and acclimate to the release site. The length of holding period varied between release sites from an unplanned immediate release (hard release) to 4–10 days (semi-soft release) to 6–16 weeks (soft-release). Field monitoring of released elk began immediately following release (February/March 1998, 1999, 2000, and 2001) and continued through to the end of the study period in December 2005. All released elk were fitted with mortality/motion-sensitive very high frequency (VHF) radiocollars (Model LMRT-4; Lotek Engineering Inc., Newmarket, Ontario, Canada), except 30 of 60 animals released in 2000 in LOW, which were not collared. Overall, the majority of marked animals were adult female (50.7%), followed by juvenile females (17.0%), juvenile males (13.6%), adult males (11.0%), yearling males (4.4%), and yearling females (3.1%). The percent of the estimated elk population on a given site that was radiocollared during the study period ranged between 25 and 97% in BNH ($n = 120$ radiocollared elk), 56 and 100% in NFR ($n = 177$ radiocollared elk), 50 and 71% in LOW ($n = 73$ radiocollared elk), and 75 and 100% in LHNS ($n = 50$ radiocollared elk). Elk restored to the BNH and LHNS were located weekly, whereas those released in NFR and LOW were

located biweekly. Both ground and aerial telemetry techniques were regularly used for survival monitoring and cause of death assessment.

Assessment of Mortality Factors

Causes of elk mortality were interpreted on the basis of animal behavior prior to death (i.e. behavior indicating meningeal [*Parelaphostrongylus tenuis*] worm infection) (Anderson & Lankester 1974) and ground-tracking and assessing the carcass as soon as possible (average 6 ± 2 days) after receiving a mortality signal. Physical evidence and condition of the remains were used to determine the cause of death. In cases where the cause of death was undetermined, a veterinary pathologist (Canadian Cooperative Wildlife Health Centre, University of Guelph) determined the cause of death via necropsy.

The proximate cause of death was classified into categories based on evidence at the death site and ancillary information from carcass necropsy. The categories included: predation, illegal hunting, accidental death (vehicle collision, injury, and drowning), emaciation, probable meningeal worm, death within 1 month of release resulting from transport injuries and/or capture myopathy, and unknown (i.e. no clear evidence of cause of death, but in most cases predation, vehicle collision, drowning, and poaching were ruled out). A total of 17 elk were excluded from this analysis as they died during capture, handling, transportation, or in holding pens, and were never released onto the Ontario landscape. The total sample was 443 elk.

Survival Estimation and Analysis

A staggered-entry Kaplan–Meier (Pollock et al. 1989) was used to estimate the study period and annual survival rates for radio-collared elk at each of the four release sites ($n = 420$). To further investigate the determinants of elk survival, Andersen–Gill hazards models were developed based on knowledge of the Ontario elk populations and published information on factors influencing elk and ungulate survival. Variables representing: (1) the four elk release sites (NFR, BNH, LOW, and LHNS); (2) time in years (1998–2005) and biological seasons: parturition and lactation (1 May–31 August), rutting period (1 September–30 November), and winter pregnancy (1 December–31 April) were included. Year and season were also included through a set of dummy variables representing year or season individually. Other variables that considered possible determinants of elk survival included (3) demographic characteristics (gender and age as dummy variables isolating adults and calves); and (4) length of holding period prior to release (<10, 11–20, 30–60, and >90 days). Two-way interaction terms among sex, age, and time variables (year of study and elk season) were also examined to see if model fit improved (Hosmer & Lemeshow 1999) (Table 1). All elk released in Ontario between 1998 and 2001 were included in this analysis ($n = 403$) with the exception of 17 animals that were thought to have died from transport-related injuries and/or capture myopathy. These animals were censored to better characterize the determinants of survival on the Ontario

Table 1. Description of variables used in analyses of elk survival in Ontario.

Variable	Description and coding system
Demographic	
Gender	Gender (female = 1)
Age class	Dummy variable representing age class (adult = 1, juvenile = 2)
Temporal	
Season	Dummy variable representing each season of the study (December–April = 1, May–August = 2, September–November = 3)
Year	Dummy variables representing each year of the study
Behavioral	
Distance from release site ^a	Annual average distance from the release site
Direction from release site ^a	Annual average direction from the release site
Annual home range size ^a	95% fixed kernel home range size (m ²)
Herd member ^a	Dummy variable representing established herd member or solitary animal (herd = 1)
Habitat	
Habitat type ^a	Proportions of seven habitat types identified by Ontario Land Sat images (water, settlement/infrastructure, sand/gravel/mine tailings/bedrock, forest depletion cuts, forest, wetlands agriculture/pasture/abandoned fields)
Anthropogenic	
Holding period	Length of holding period prior to release (<10 days = 1, 11–20 days = 2, 30–60 days = 3, >90 days = 4)
Release site	Dummy variable representing the four elk release sites in Ontario (BNH = 1, NFR = 2, LOW = 3, LHNS = 4)
Feeding sites ^a	Dummy variable representing elk in close proximity to feeding sites (feeding = 1); proxy measure for <i>Parelaphostrongylus tenuis</i> exposure

The first set of models included all elk restored to release sites in Ontario, with release site as a categorical variable. The second analysis included more detailed variables reflecting behavioral characteristics of elk restored to the BNH region of Ontario.

^aAvailable only for elk released in the BNH region of Ontario (i.e. model population).

landscape in the absence of the effects of capture and transportation.

A second series of models that included a more detailed set of variables complete for 116 elk released in the BNH region (four elk were censored as they died prior to release) were also examined. Models included all variables described above, as well as spatially explicit behavioral and environmental variables including maximum annual distance from release site, average annual direction from release site, size of annual 95% fixed kernel home range, herd membership representing established herd member or solitary animal (based on radio-telemetry and observation), habitat variables (water, human settlement, sand/gravel/mine tailings/bedrock, forest depletion cuts, forest, wetlands agriculture/pasture/abandoned fields) associated with the annual 95% fixed kernel home range estimate for each animal, and average annual proximity (in km) to known elk and white-tailed-deer feeding locations (proxy measure for *P. tenuis* exposure) (Table 1). Habitat variables were derived from the Ontario Land Cover database (Spectranalysis Version 2000). ArcView 3.2 was used for all Geographic Information System analyses, including estimation of 95% fixed kernel home ranges for elk, which were calculated using the Home Range Extension for ArcView. Two-way interaction among sex, age, and time variables in the model (year of study and season) were also examined to see if model fit improved (Hosmer & Lemeshow 1999).

An a priori suite of candidate models for all elk released in Ontario and for elk released in the BNH region was developed. For each of the models, determinants of elk survival were assessed using Andersen–Gill hazard models (Andersen et al.

1993). To avoid problems associated with multicollinearity, we analyzed tolerance and variance inflation factors (VIF; Tabacknick & Fidell 2001). Tolerance statistics ranged from 0.447 to 1.326, and VIF ranged from 2.213 to 5.287 for all variables under consideration; therefore, we considered multicollinearity acceptably low and did not eliminate any variables from the analysis.

Andersen–Gill models are based on the assumption of proportional hazards. To test this assumption, scaled Schoenfeld residuals versus survival times were plotted, and a chi-square significance test was examined for the resulting correlation (Therneau & Grambsch 2000). The assumption of proportional hazards was met for all models.

Andersen–Gill models were compared using standard model selection and multimodel inferencing procedures. If more than one model was considered plausible, model averaging techniques were used to more accurately include model selection uncertainty of the main effects directly into the elk hazards. The relative importance of each individual variable was also examined by summing AIC weights (w_i) across all models that contained the variable of interest (Burnham & Anderson 2002).

Results

Mortality

A total of 218 mortalities were documented for 443 elk released across the four Ontario release areas between February/March 1998 and December 2005. Overall mortality for all elk released in Ontario was 49%, ranging from a low of 19% in the LHNS

Table 2. Percent cause-specific mortality for elk restored to four sites in Ontario, Canada, including NFR, BNH, LOW, and LHNS, from January 1998 to December 2005.

Release site	Release year	Total mortality	Transport injuries/ capture myopathy	Emaciation	Predation	Poaching	Accident	<i>P. tenuis</i>	Unknown
NFR	1998	70 (28/40)	46 (13)	0	35 (9)	4 (1)	12 (3)	0	8 (2)
NFR	1999	85 (56/66)	23 (13)	32 (14)	18 (18)	0	16 (7)	0	9 (4)
NFR	2000	54 (19/35)	0	5 (1)	58 (11)	0	37 (7)	0	0
NFR	2001	57 (13/23)	15 (2)	0	46 (6)	0	38 (5)	0	0
BNH	2000	53 (37/70)	22 (8)	13 (4)	6 (2)	16 (5)	26 (8)	3 (1)	29 (9)
BNH	2001	46 (23/50)	0	0	9 (2)	26 (6)	13 (3)	13 (3)	39 (9)
LOW	2000	50 (15/30)	0	0	20 (3)	27 (4)	20 (3)	0	33 (5)
LOW	2001	42 (18/43)	33 (6)	0	18 (3)	6 (1)	18 (3)	0	29 (5)
LHNS	2001	19 (9/47)	11 (1)	0	0	22 (2)	22 (2)	0	44 (4)
All	1998–2001	49 (218/443)	20 (43)	9 (19)	25 (54)	9 (19)	19 (41)	2 (4)	17 (38)

Sample sizes are given in parentheses. Total mortality is accumulated mortality for radiocollared elk from release year to 31 December 2005. Elk that died prior to release (i.e. at EINP, in transport, in holding pens) were eliminated from the analyses ($n = 17$).

region to a high of 85% in the NFR region (Table 2). Predation by wolves and possibly black bears was the most important proximate cause of mortality (25%), largely owing to comparatively high mortality in the NFR region where 38% of all mortalities were attributed to predation. No mortalities in the LHNS region were attributed to predation, whereas only 3 and 8% of mortalities were attributed to predation in the BNH and LOW regions, respectively. Death within 1 month of release due to injuries sustained during translocation and/or post-capture myopathy was the cause of death for 20% of all elk in Ontario, followed by accidents, including vehicle collision, injury and drowning, (19%), emaciation (9%), illegal hunting (9%), and infection with the parasite *Parelaphostrongylus tenuis* (2%). An additional 17% of elk died of unknown causes.

Survival

Survival of 420 individual radiocollared elk was monitored across the four Ontario release areas for an average of 899 ± 13 (SE) days per animal. During the study period (1998–2005), the survival rate for all elk was 0.51 (0.41–0.62), whereas across each of the four release areas survival ranged from 0.45 (0.37–0.53) in NFR, 0.54 (0.45–0.63) in BNH, 0.62 (0.49–0.782) in LOW, and 0.81 (0.66–0.90) in LHNS (Fig. 2). Annual survival rates for all released elk were highest in the final year of this study, 2005, with a small but steady increase observed during the years 2003–2005. Annual survival rates for elk released in the NFR regions ranged from a high of 0.88 (0.60–0.98) in 2005 to a low of 0.49 (0.39–0.59) in 2001, whereas for elk released in the BNH region annual survival rates ranged from a high of 0.80 (0.69–0.88) in 2005 to a low of 0.59 (0.48–0.68) in 2002. For elk released in the LOW region, annual survival rates ranged from a high of 0.87 (0.45–0.98) in 2005 to 0.57 (0.42–0.69) in 2002, whereas for those elk released in LHNS, annual survival rates ranged from a high of 0.98 (0.95–0.99) in 2005 to a low of 0.80 (0.66–0.89) in 2003.

Andersen–Gill hazard models for all elk released in Ontario (excluding censors, total $n = 403$) revealed six candidates with substantial support ($\Delta_i < 4$) (Table 3). These top six models represented 0.96 of the available model weight and

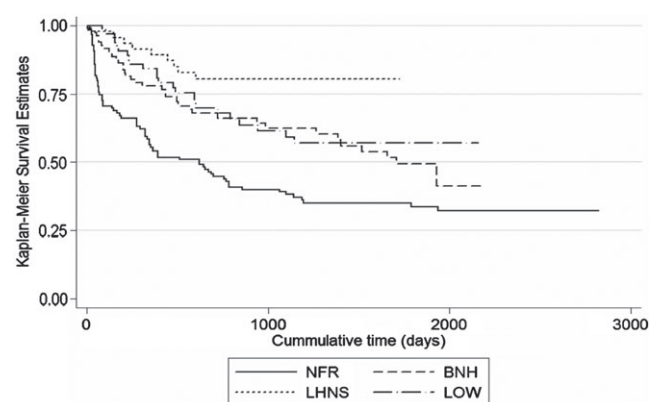


Figure 2. Kaplan–Meier survival rates for elk restored to four release sites in Ontario including NFR, BNH, LOW, and LHNS from 1998 to 2001. Cumulative time 0 represents the release day into the study rather than calendar time.

variables representing length of holding time ($\sum w_i = 0.822$) and year of study ($\sum w_i = 0.797$) had the most collective weight of evidence, followed by season ($\sum w_i = 0.766$), sex ($\sum w_i = 0.295$), age ($\sum w_i = 0.181$), and the interaction term age $\times$ year ($\sum w_i = 0.173$). Because our analysis revealed six highly plausible models (i.e. $\Delta_i < 4$), model-averaged hazard estimates indicated that animals subject to limited holding (< 10 days) prior to release were 23.3 (95% confidence interval [CI]: 20.404–26.077) times less likely to survive than those who were held longer prior to release (> 11 days) (Table 4). Year of study, elk season, sex, age, and the interaction term age $\times$ year were also associated with higher elk hazards; however, these variables tended to have a smaller effect on elk mortality risk. More specifically, model-averaged hazard estimates indicated that elk were less likely to survive in the year immediately following release (0.478; 95% CI: 0.359–0.659). Elk were also less likely to survive during the winter months December–April (0.591; 95% CI: 0.358–0.822). Sex and age also influenced survival with male elk (0.240; 95% CI: 0.035–0.444) and calves (0.769; 95% CI: 0.726–0.810) experiencing higher rates of survival. Finally, the interaction term of age $\times$ year revealed

Table 3. Candidate models for elk survival in four release sites in Ontario, Canada December 1999 to December 2005.

	<i>Model description</i>	<i>K</i>	<i>AIC</i>	Δ_i	w_i	χ^2	<i>P</i>
1	Season + year of study + holding time	4	1420.15	0	0.290	78.05	<0.001
2	Year of study + holding time	3	1420.60	0.45	0.231	75.60	<0.001
3	Sex + age + elk season + year of study + age $\times$ year	5	1420.62	0.48	0.176	78.39	<0.001
4	Season + year of study	3	1421.15	1.00	0.169	75.04	<0.001
5	Sex + season + year of study + holding time	5	1421.23	1.08	0.123	78.97	<0.001
6	Sex + age + season + year of study + holding time	6	1421.86	1.71	0.033	80.34	<0.001
7	Age + season + release area + year of study + holding time	6	1424.52	4.37	0.026	81.68	0.0023
8	Sex + age + season + release area + year of study + holding time	7	1424.98	4.83	0.010	83.22	0.0047

For each Andersen–Gill model, we include number of parameters (*K*), Akaike's Information Criterion (*AIC*), *AIC* difference (Δ_i), *AIC* weight (w_i), and goodness-of-fit (*P*) ($n = 404$). Only models with $<5\Delta_i$ are provided.

Table 4. Model averaged hazards ratios and unconditional standard errors for the top 6 ($<4\Delta_i$) models developed to explain survival of elk in Ontario, Canada, December 1999–2005.

<i>Parameter</i>	<i>Hazard ratio</i>	<i>SE</i>	95% <i>CI</i> <i>Upper</i>	<i>Lower</i>
Season	0.591	0.119	0.822	0.358
Year of study	0.478	0.061	0.598	0.359
Holding time	23.248	1.451	26.077	20.404
Sex	0.240	0.105	0.444	0.035
Age	0.769	0.021	0.810	0.726
Age $\times$ year	1.835	0.540	3.267	0.030

that adult elk experienced lower rates of survival in the year immediately following release (1.835; 95% CI: 0.030–3.267) (Table 4).

To further investigate the factors influencing elk survival, a second set of models was developed using behavioral and spatial variables that were complete for animals released in BNH ($n = 112$). Six candidate models were considered to have substantial support ($\Delta_i < 4$), and included variables representing size of annual home range ($\sum w_i = 0.998$) and direction from release site ($\sum w_i = 0.922$), followed by year of study ($\sum w_i = 0.722$), percentage of agricultural land in the home range ($\sum w_i = 0.559$), percentage of human settlement in the home range ($\sum w_i = 0.505$), interaction term age $\times$ year ($\sum w_i = 0.447$), and the interaction term sex $\times$ age ($\sum w_i = 0.371$) (Table 5). More specifically, model-averaged estimates of elk hazards for the top six models revealed that elk with comparatively large annual home ranges were qualitatively (1.09; CI: 0.648–1.132) less likely to survive than those with smaller home ranges (Table 6). As well, elk whose average direction was south/southwest of the release site were qualitatively 1.001 (95% CI: 0.638–1.336) times less likely to survive than those that dispersed in other directions.

Year of study, percentage of agricultural land in the home range, percentage of human settlement in the home range, and interaction terms age $\times$ year and sex $\times$ age were also associated with higher elk hazards; however, these variables had a smaller effect on elk survival and a lower weight of evidence. Notably, elk were less likely to survive in the year immediately following release (0.490; 95% CI: 0.306–0.674). The risk of mortality was also higher for elk whose home range consisted of higher proportions of agricultural land (0.548; 95% CI: 0.211–0.886)

and human settlement (0.187; 95% CI: 0.011–0.364). Finally, the interaction terms age $\times$ year (0.919; 95% CI: 0.554–1.522) and sex $\times$ age (0.899; 95% CI: 0.333–2.421) were also associated with higher elk hazards, with adult female elk experiencing lower rates of survival, especially in the year immediately following release (Table 6).

Discussion

Overall, this study revealed that survival of elk was low compared with established populations, but was similar to rates reported for reintroduced or colonizing populations (Raedeke et al. 2002; Larkin et al. 2003). As we anticipated, the lowest annual survival rates were generally recorded in the first year immediately following release, largely owing to death from capture myopathy and/or transport-related injuries; however, factors associated with inexperience or unfamiliarity with the Ontario landscape, such as high rates of predation, accidental death, and poaching were also important causes of death in the years immediately following release. Once elk had been on the landscape for 2–3 years, annual mortality tended to decrease. These results are similar to those reported for elk reintroduced to Kentucky (U.S.A.), where 49% of all mortalities were attributed to capture injuries (Larkin et al. 2003), with the remaining related to accidents and movement outside the restoration zone.

Analyses of the mechanisms responsible for elk survival in Ontario garnered further support for the hypothesis that survival rates increase as an animal's time and experience on the landscape increase. Indeed, both holding time and year of study were strong predictors of elk survival in Ontario. Increasingly, the method of release is recognized as an important factor in

Table 5. Candidate models for elk survival in the BNH region of Ontario, Canada December 1999 to December 2005.

Model description	<i>K</i>	<i>AIC</i>	Δ_i	w_i	X^2	<i>P</i>
1 Year of study + direction from release site + agriculture + home range size	5	146.011	0.000	0.386	25.28	<0.001
2 Year of study + direction from release site + settlement + home range size	5	147.818	1.806	0.157	24.42	<0.001
3 Year of study + direction from release site + settlement + agriculture + home range size	6	148.153	2.142	0.132	26.19	<0.001
4 Year of study + direction from release site + home range size	4	148.933	2.922	0.090	21.41	<0.001
5 Year of study + direction from release site + agriculture + home range size + age $\times$ year	6	148.959	2.939	0.089	25.39	0.001
6 Year of study + direction from release site + agriculture + home range size + sex $\times$ age	6	149.012	3.001	0.086	25.43	0.001

For each Andersen–Gill model, we include number of parameters (*K*), Akaike's Information Criterion (*AIC*), *AIC* difference (Δ_i), *AIC* weight (w_i), and goodness-of-fit (*P*) ($n = 116$). Only models with $<5\Delta_i$ are provided.

Table 6. Model averaged hazards ratios and unconditional variances for the top 6 ($<4\Delta_i$) models developed to explain survival of elk released in the BNH region of Ontario, Canada, January 2000–December 2005.

Parameter	Hazard ratio	SE	95% CI Upper	Lower
Year of study	0.490	0.094	0.674	0.306
Direction from release site	1.001	0.178	1.336	0.638
Agriculture	0.548	0.173	0.886	0.211
Settlement	0.187	0.091	0.364	0.011
Home range size	1.09	0.181	1.132	0.648
Age $\times$ year	0.919	0.237	1.522	0.554
Sex $\times$ age	0.899	0.455	2.421	0.333

the success of wildlife restoration programs, with soft release involving holding animals through an acclimation period being beneficial (Haydon et al. 2008; Yott et al. 2010).

In Ontario, the length of time that elk were held prior to release varied from 16 weeks in the LHNS region to no acclimation in the BNH region, following an escape/release from the enclosure the day they were placed in the holding pen. The relationship between holding time and survival for Ontario elk may be related to increasing site fidelity and social cohesion for animals experiencing a soft release. For a recently restored species, a balance of dispersal and coalescence generally determines the distribution of animals across the landscape (Haydon et al. 2008), and elk survival is generally higher for animals in groups (Raedeke et al. 2002). Therefore, survival may, in part, depend on the coalescence process limiting widespread dispersal across the landscape and the establishment of an effective group structure within the release area (Haydon et al. 2008).

For elk released in the BNH region of Ontario, hazards increased with size of the animals' home range. Following release in a novel environment, animals often experience a dispersive phase in which they rapidly emigrate from the point of origin, possibly to explore the novel landscape and to find good quality habitat (Haydon et al. 2008). Dispersion may also be influenced by the proximity of other animals, as was the case with attraction of recently released animals to elk farms in the LOW, BNH, and LHNS regions (Ryckman et al. 2007). Alternatively, elk may have dispersed to more suitable habitat and high-quality forage conferring lower risk, although this scenario is unlikely because the BNH elk release site was located in the core of the historic elk range in Ontario, and habitat quality and forage availability were considered good based on a recent elk habitat suitability index (Hutchinson et al.

1997). The increase in hazards with the size of an animals' home range may be related to a lack of group structure and social cohesion (Haydon et al. 2008), as well as less suitable habitat available farther from the carefully selected release sites (Rosatte et al. 2007).

In this study, elk that moved in a southward direction also experienced qualitatively lower survival than those that moved in a more northerly direction. Owing to human population growth and associated development, elk habitat availability was restricted in southern Ontario, the core of their historic range. Southward movements may reflect searching for better quality habitat unavailable at the chosen release sites. Although southern regions may provide better elk habitat than some of the actual release areas, including earlier spring green-up and high-quality forage (Hutchinson et al. 1997), attendant increase in proximity to anthropogenic activities likely increased mortality caused by humans. Indeed, the highest rates of vehicle collisions and illegal hunting were recorded in the BNH region—the most southerly and most densely-populated region of the current elk range in Ontario (Hutchinson et al. 1997). Studies elsewhere also indicate that human-caused mortality is generally important for recovering ungulate populations, especially those with low predator densities (Larkin et al. 2003).

Finally, we anticipated that elk located in release sites closer to the core of the historic range of elk in Ontario would have higher survival, as habitat quality and predator density would be high and low, respectively (Hutchinson et al. 1997). Although elk in the NFR and BNH regions had slightly lower overall survival rates compared with those in the LOW and LHNS regions, release site was not a predictor of survival. These results are unexpected given the disparity in habitat features and locations of the sites (Hutchinson et al. 1997).

Our results may be explained by the fact that although the main causes of mortality differed for each release site (i.e. predation was the leading cause of death for elk released in the NFR and LOW release area, whereas vehicle collisions and poaching were the leading causes of death for elk in the BNH region), overall mortality rates were similar across all four release sites.

Implications for Practice

- Capture myopathy and transportation-related injuries can be an important cause of mortality during the initial stage of a wildlife restoration project and should be considered a predictable factor to incorporate into planning.
- Survival of elk is largely influenced by the method of introduction to the novel landscape and behavior in the first year following release. Actions such as holding animals in an enclosure prior to release can mitigate disruption of natural group structure and provide acclimation and familiarity to the novel environment.
- The negative effect of proximity to humans implies that release sites further from human settlement, coupled with efforts to minimize dispersal away from the carefully chosen release site, can improve the chances of population persistence.

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